

Multiple Access Protocols-Part V

Comprehensive Course on Computer Networks

CS & IT Engineering

Computer Network
Medium Access Control



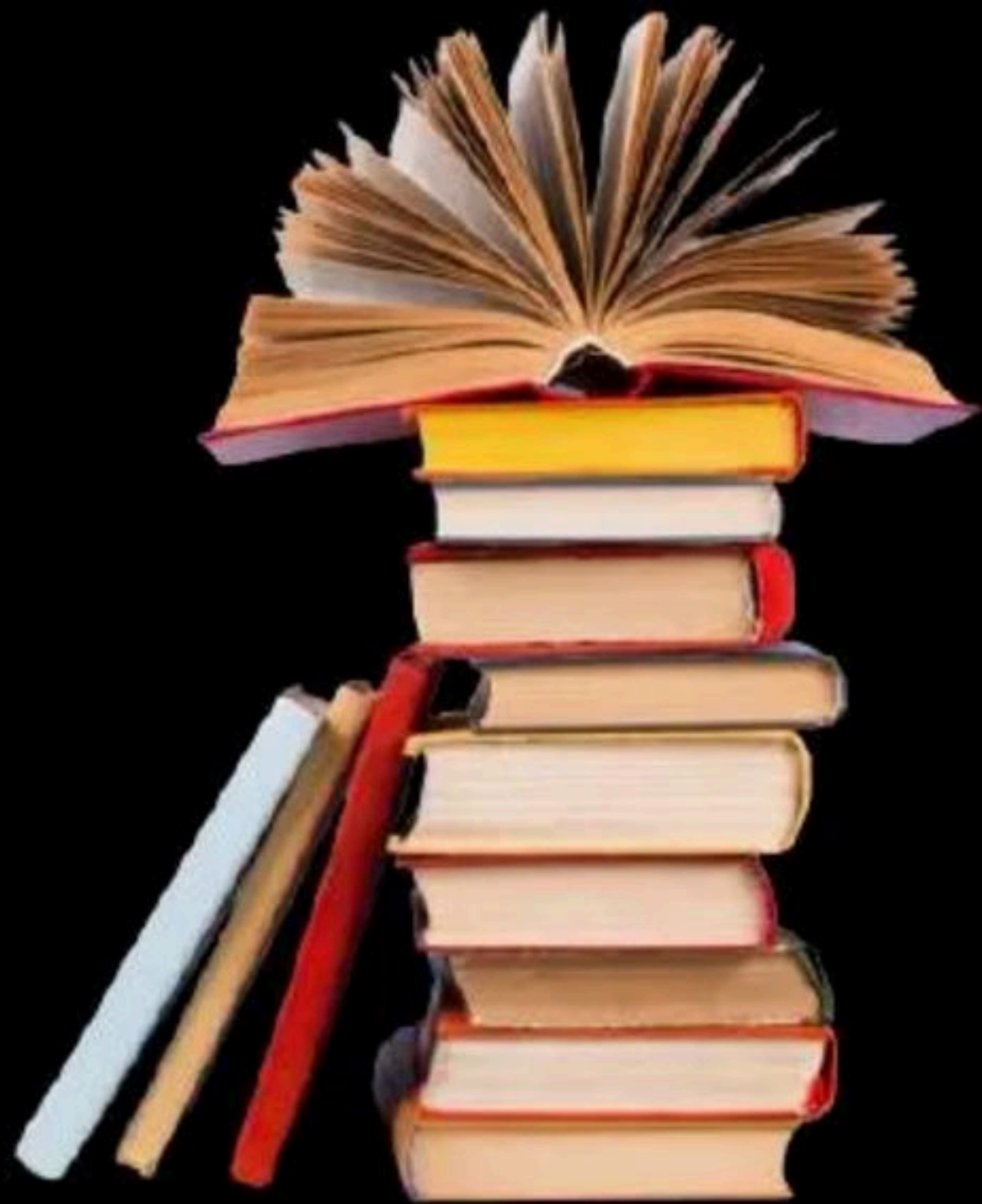
Lecture Number

By- Ankit Sir



Topics

to be covered



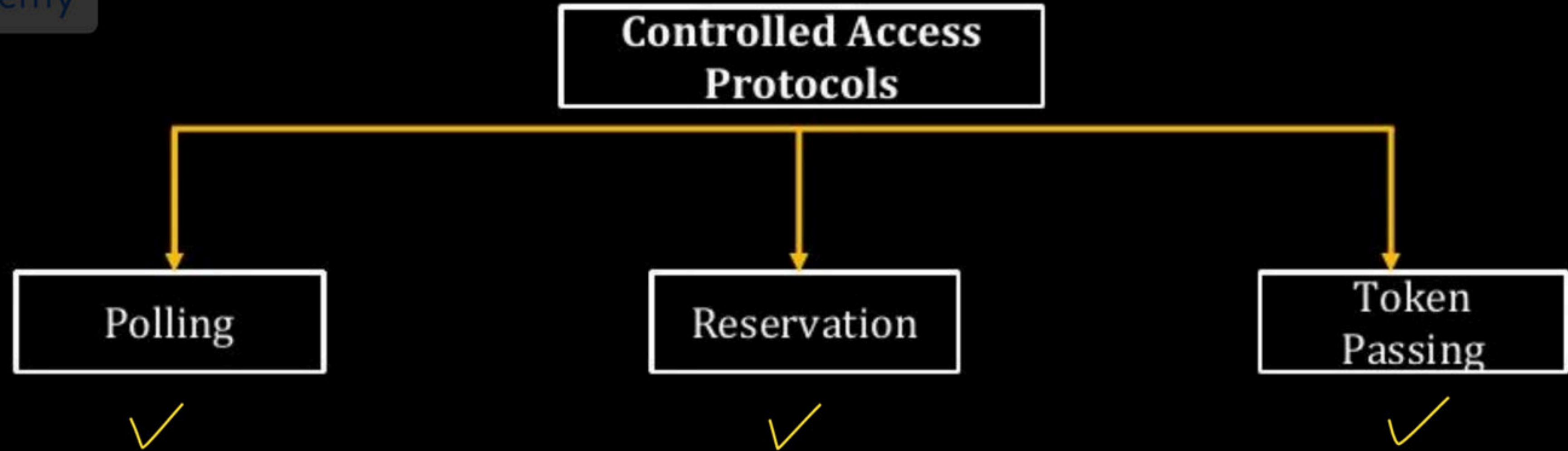
Controlled Access Protocols

Controlled Access Protocol

It can be done in the following two ways

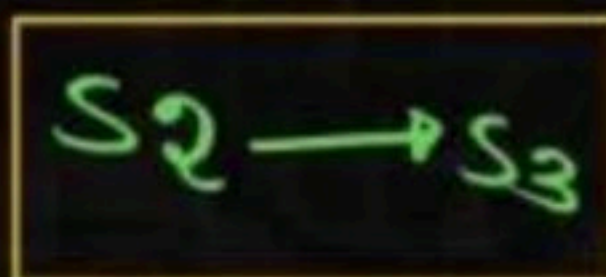
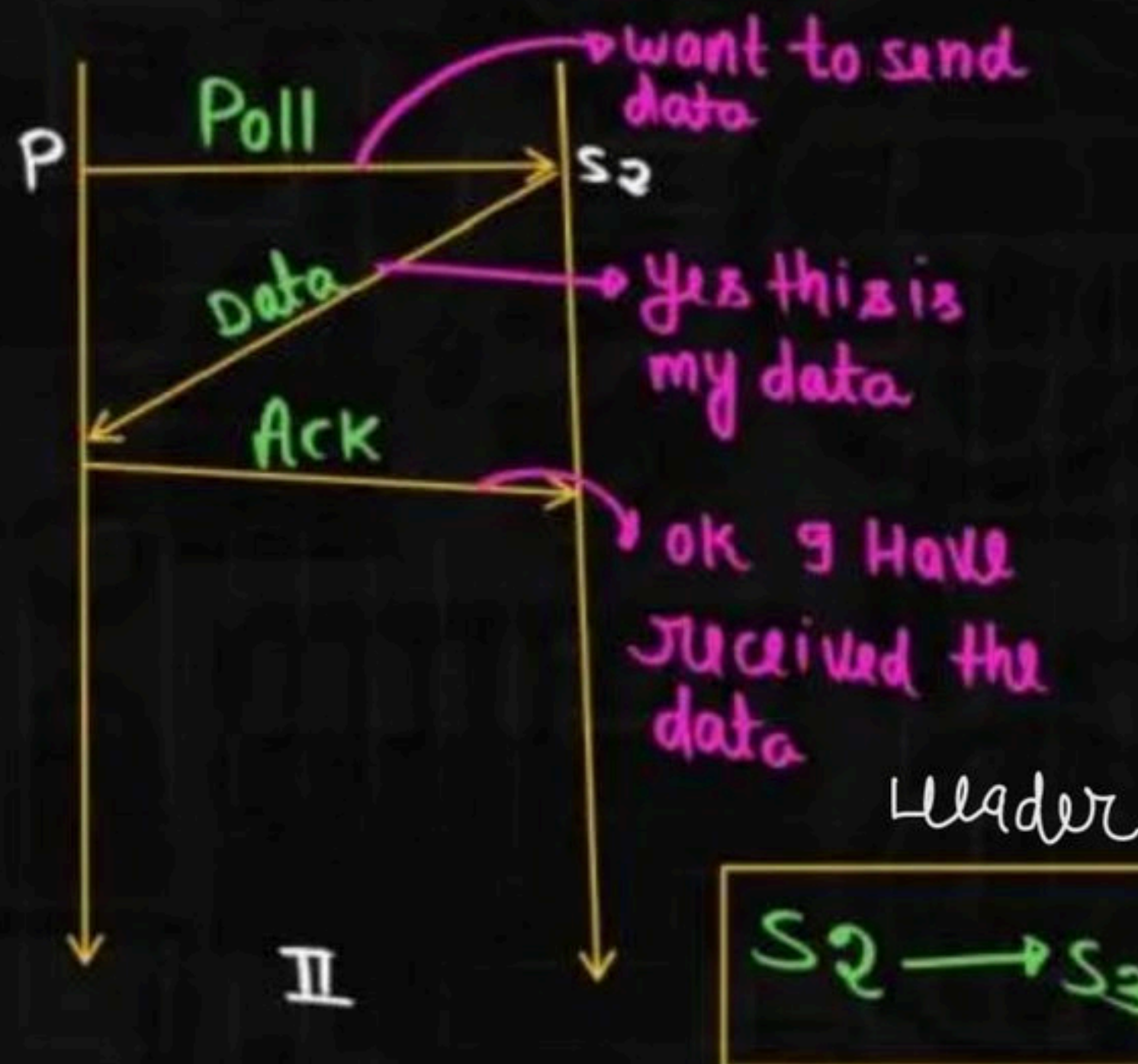
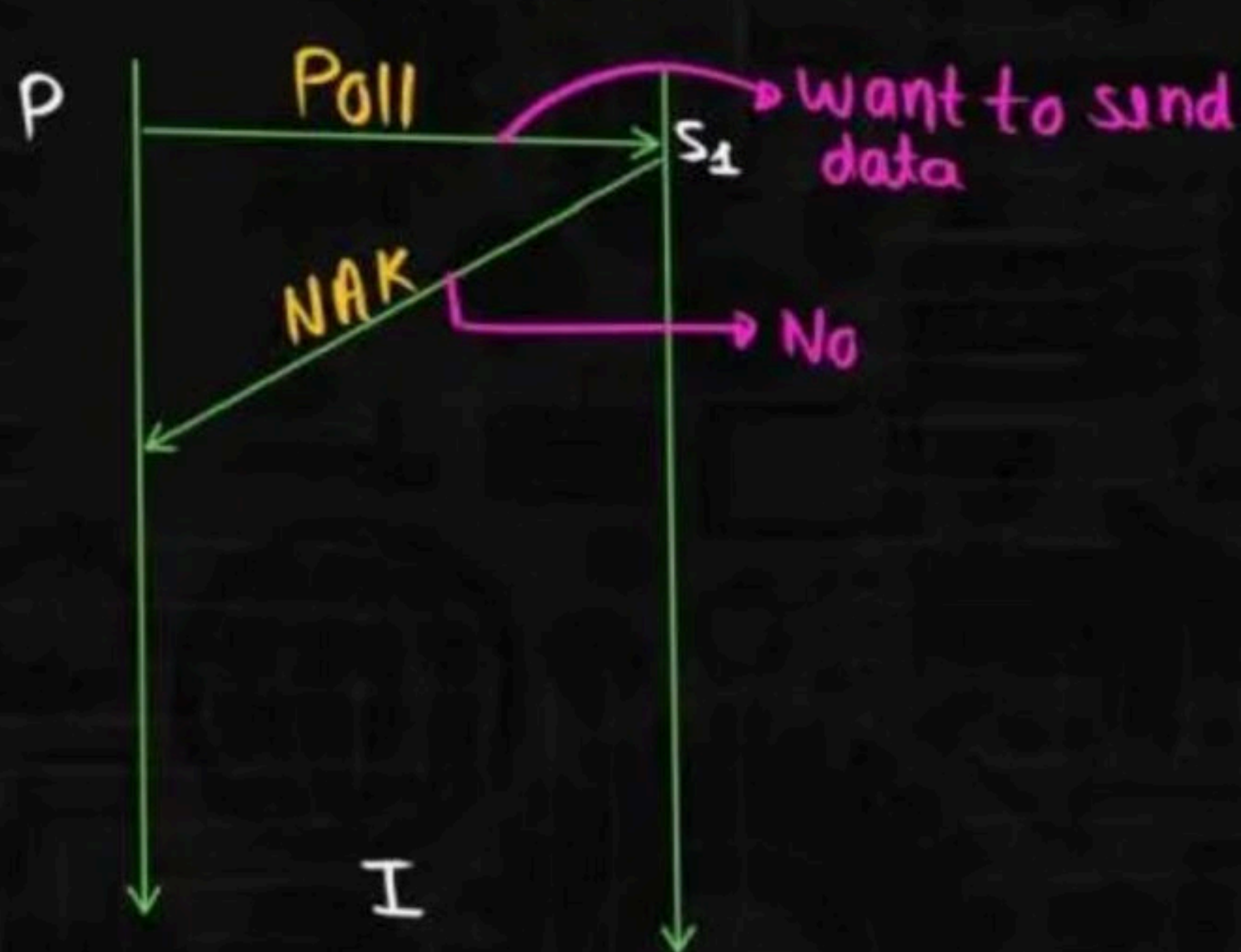
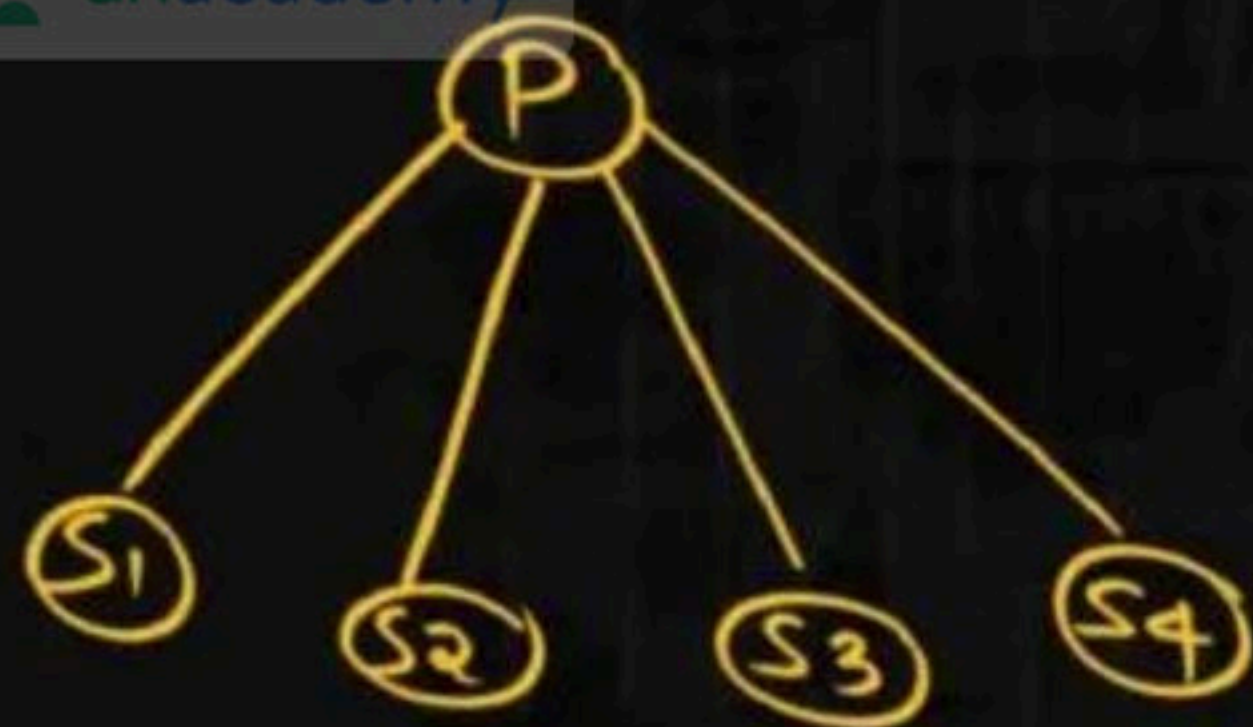
1. We have a concept of primary and secondary station. Primary station will control all secondary station.
2. If concept of primary or secondary station is missing then if any station want to send the data , it can send only in a case if all other stations gives permission to it.

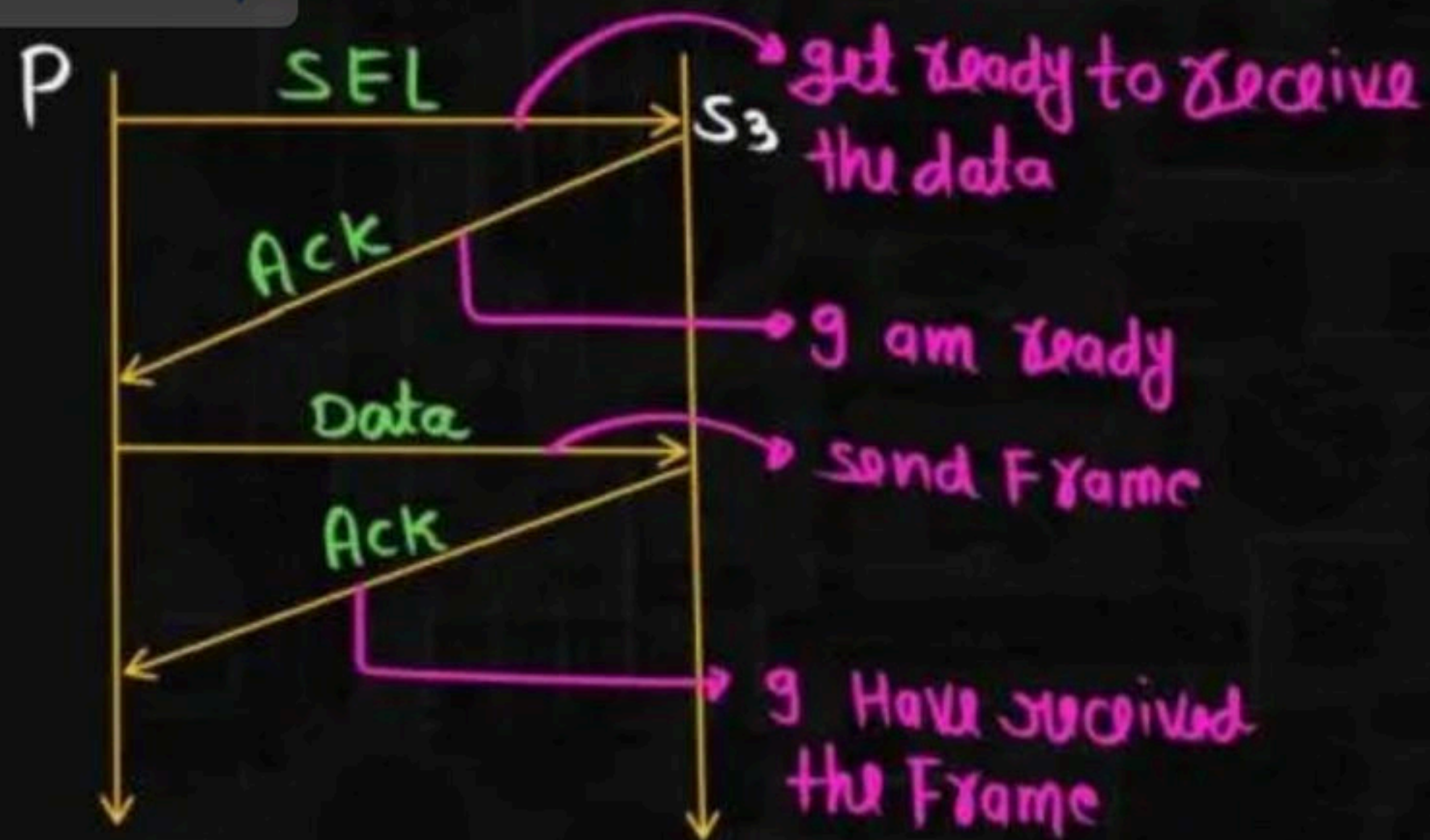
Note: controlled Access Protocol are collision free protocol.



Polling



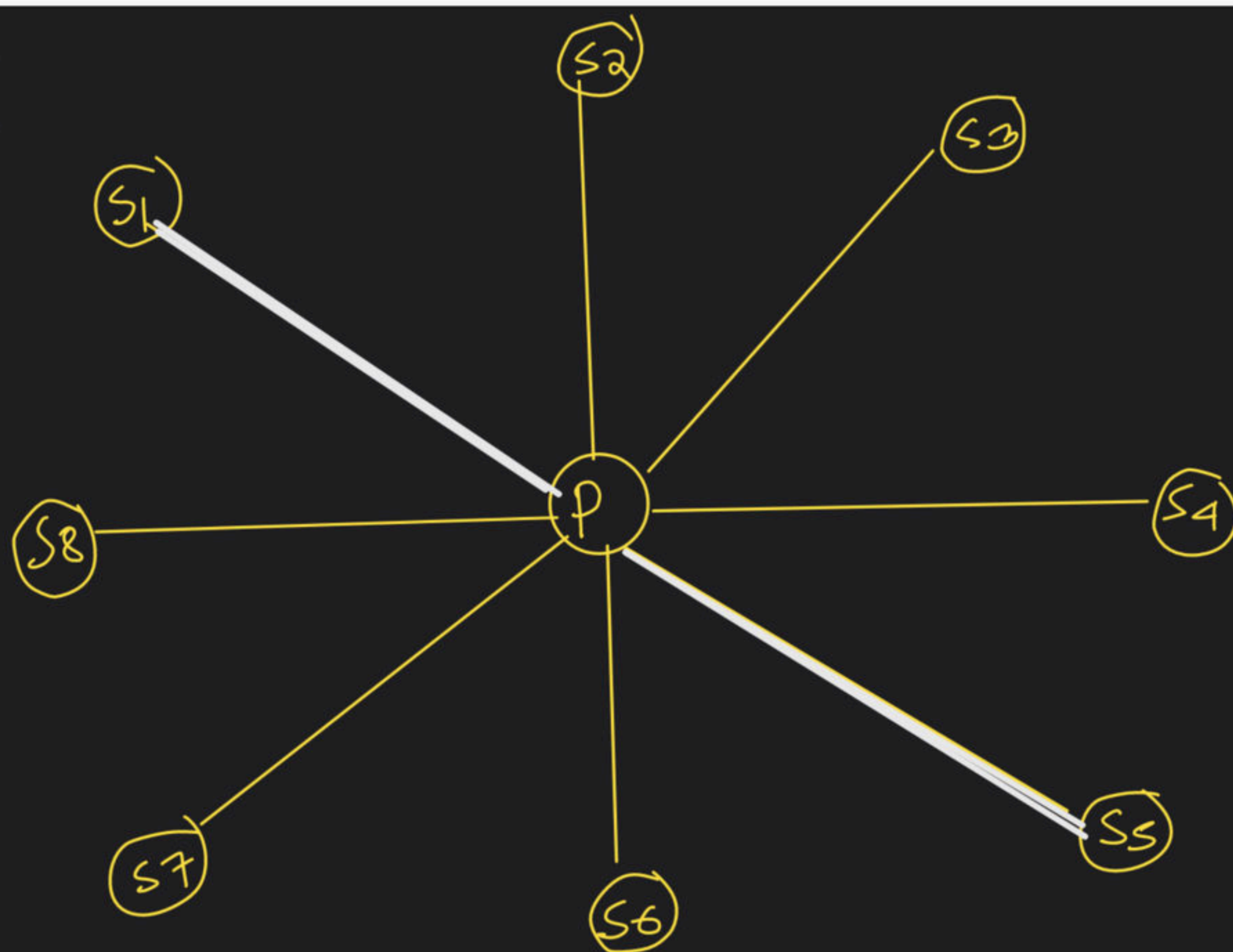




III

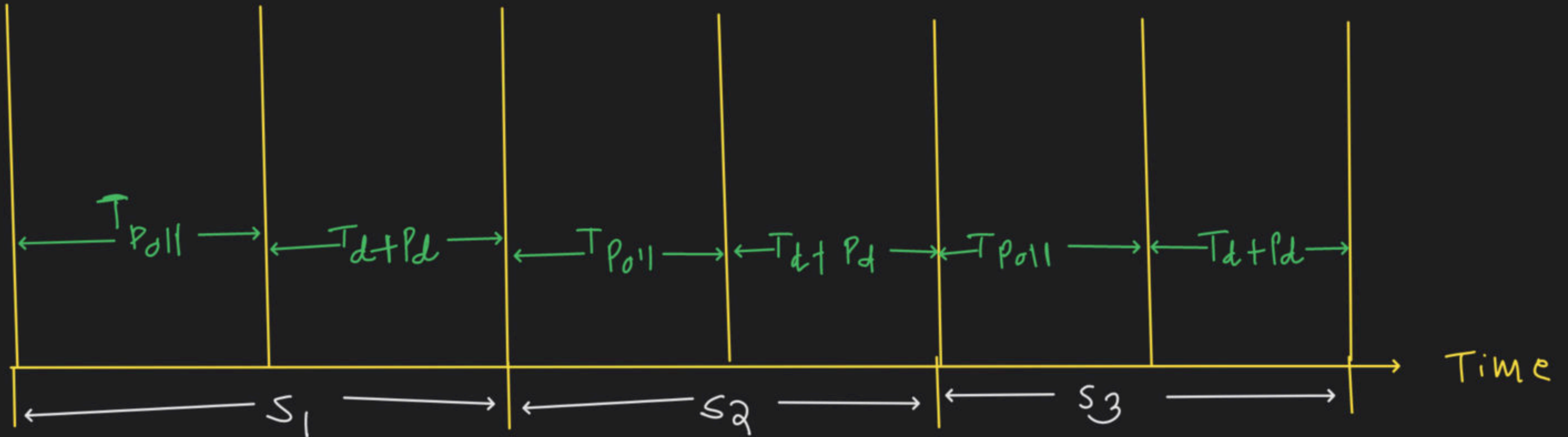


- Every station will send the data through primary station.
- Two secondary station can not communicate directly.
- Common topology used in polling mechanism is star topology.
- Bandwidth utilization is low because lots of time is wasted by sending the poll message.
- Drawback of polling is if primary station fails ,the system goes down.



$$S_1 \longrightarrow S_5$$

efficiency OF Polling



$$\text{efficiency} = \frac{\text{Useful time}}{\text{total time}} = \frac{T_d(F)}{T_{poll} + T_d(F) + P_d}$$

$$\text{Throughput} = \text{efficiency} \times \text{Bandwidth}$$

Q.1

A broadcast channel has 10 nodes and total capacity of 10 Mbps. It uses polling for medium access. Once a node finishes transmission, there is a polling delay of $80 \mu s$ to poll the next node. Whenever a node is polled, it is allowed to transmit a maximum of 1000 bytes. The maximum throughput of the broadcast channel is:

[GATE - 2005]

A. 1Mbps

☒ B. 100/11Mbps

C. 10Mbps

D. 100Mbps

$$B = 10 \times 10^6 \text{ bits/sec}$$

$$T_{\text{Poll}} = 80 \mu\text{sec}$$

$$\text{Packet size} = 1000 \text{ byte} = 8000 \text{ bits}$$

$$T_d(F) = \frac{8000 \text{ bits}}{10 \times 10^6 \text{ bits/sec}} = 800 \times 10^{-6} \text{ sec} = 800 \mu\text{sec}$$

$$\text{Throughput} = \eta * B$$

$$= \frac{10}{11} * 10 \text{ Mbps}$$

$$= \frac{100}{11} \text{ Mbps}$$

$$\text{efficiency} = \frac{\text{Useful time}}{\text{total time}}$$

$$= \frac{T_d(F)}{T_{Poll} + T_d(F) + \cancel{P_d}}$$

$\swarrow$
0

$$= \frac{800 \text{ } \cancel{\mu\text{s}}}{80 + 800 \text{ } \cancel{\mu\text{s}}}$$

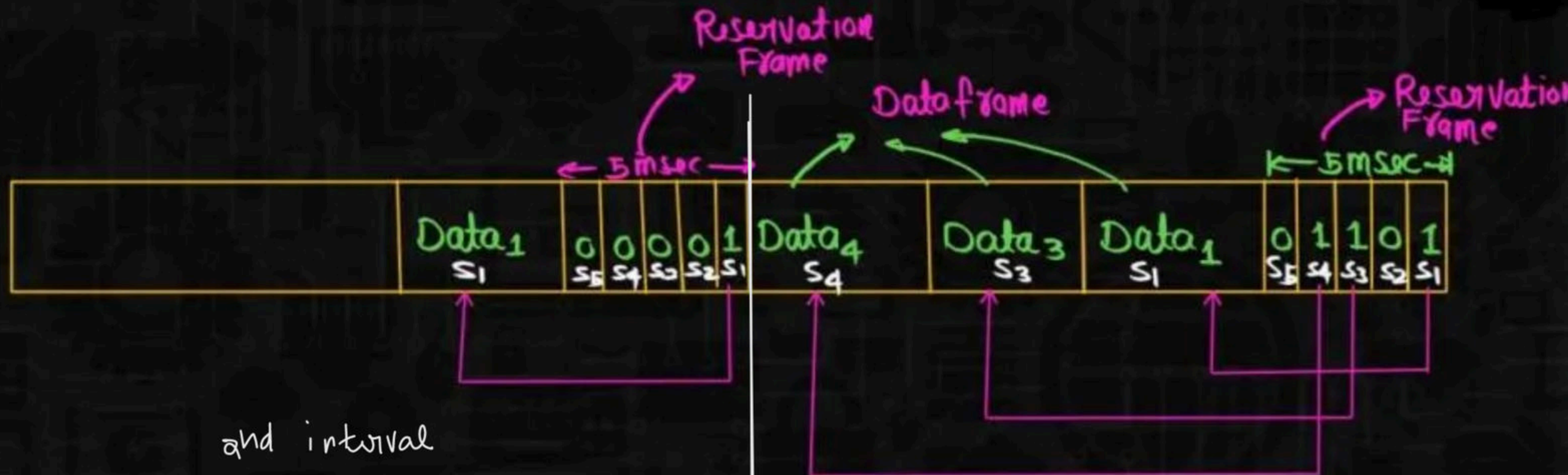
$$= \frac{800}{880}$$

$$= \frac{10}{11}$$

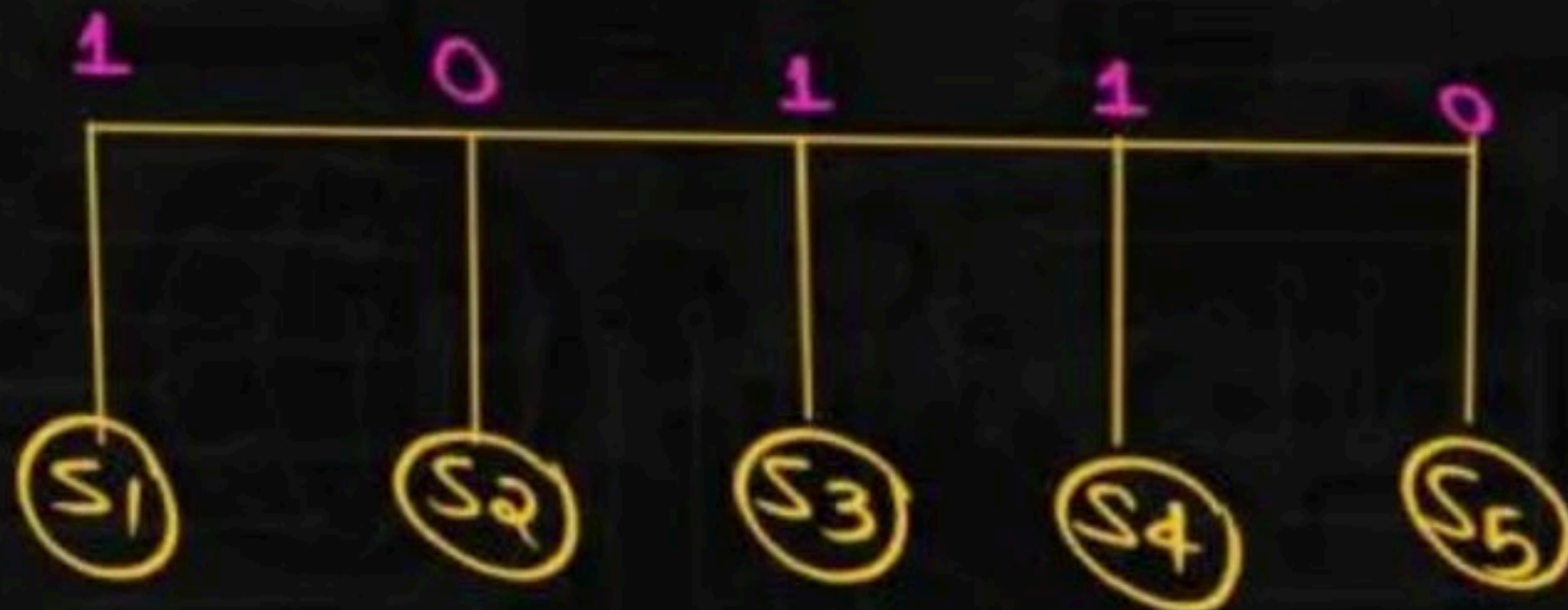
Reservation



1. In the reservation method , a station needs to make a reservation before sending the data.
2. Time is divided into intervals.
3. In each interval reservation frame precedes the data frames sent in that interval.
4. If there are N- stations in the system ,there are exactly N reservation minislots in the reservation frame. Means every station have its own minislot.
5. The stations that have made reservations can send their data frames after the reservation frame.



and interval



1st interval

Example

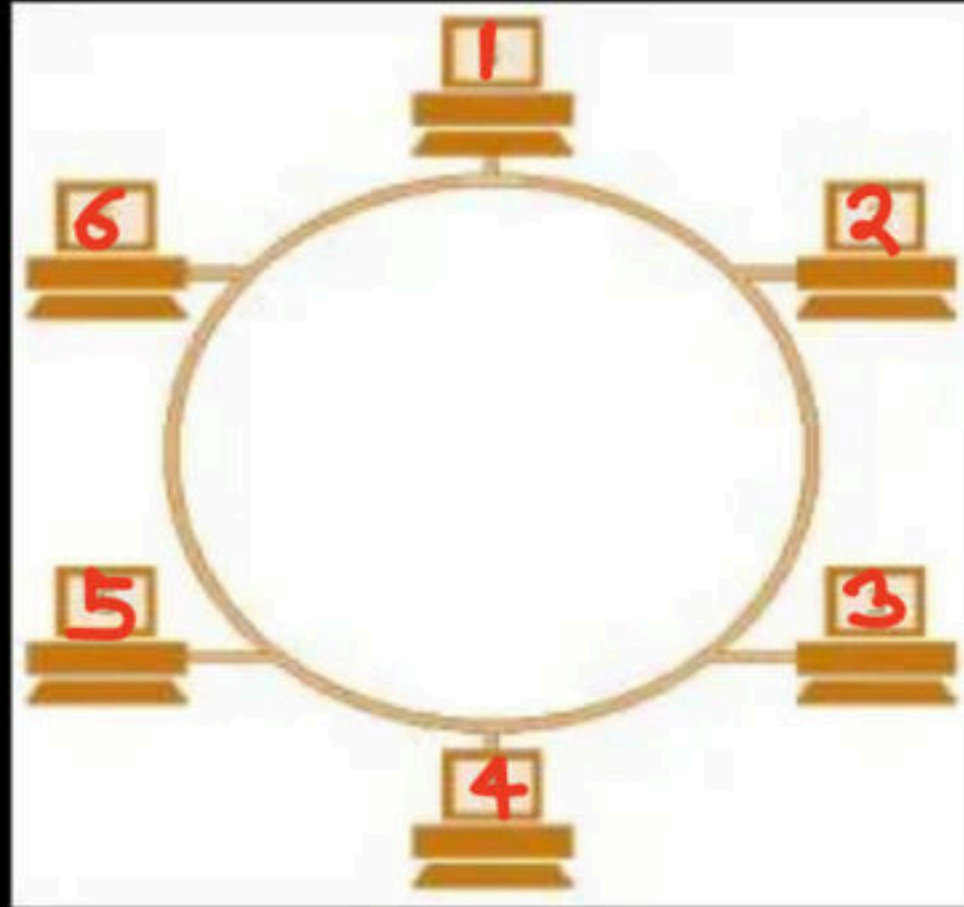
- Let us take an example of 5 stations and 5 mini slots in the reservation frame. In the first interval only 1, 3 and 4 have made reservations. In the second interval, only station 1 has made reservation.



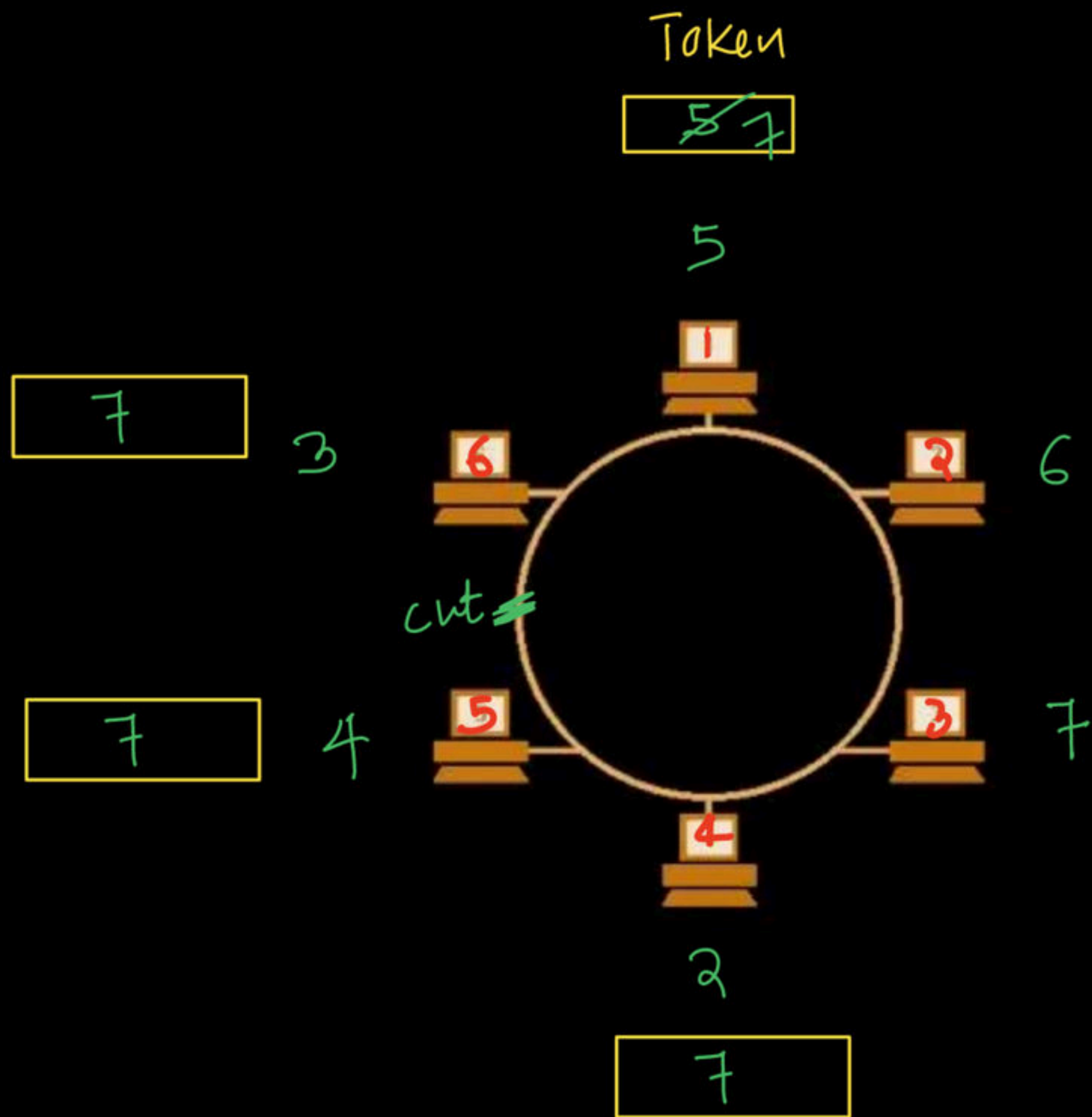
Token Passing

Token Passing

1. All the stations are logically connected to each other in the form of ring.



2. It uses a special frame called “token” that travels around the ring.
3. A station is allowed to transfer the data packet if and only if it has token.
4. Whenever station has no more data to send , it release the token.



Priority = 3 bit

Range

0 to $2^3 - 1$

0 to 7

5, 6, 7

5, 7, 2

S3 → will transfer 1st

Note:

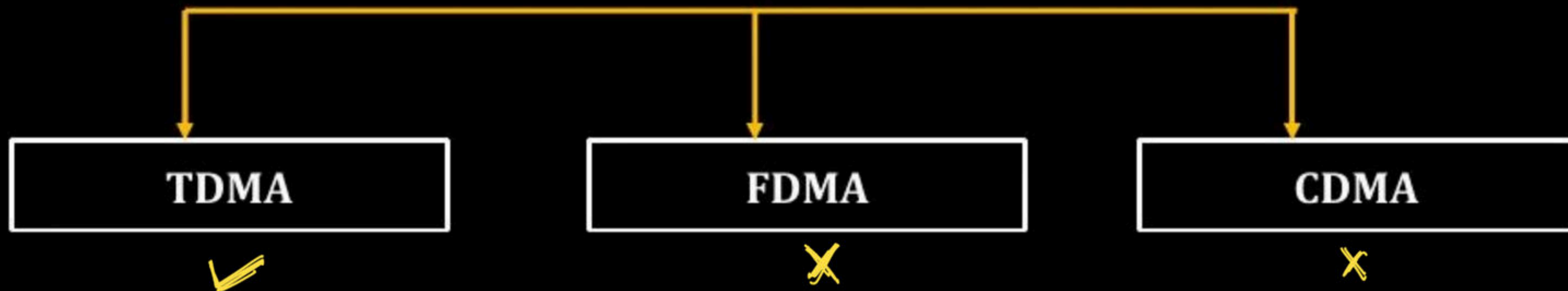
- Best technique for broadcasting the packet.
- No acknowledgement.



Channelization Protocols



Centralized Access Protocols





TDMA(Time division Multiple Access)

In time division multiple access-

- Time of the link is divided into fixed size intervals called as time slots.
- Time slots are allocated to the stations in round robin manner.
- Each station transmit its data during the time slot allocated to it.
- In case, station does not have any data to send, its time slot goes waste.

Disadvantage

- If any station does not have data to send during its time slot, then its time slot goes waste.
- This reduce the efficiency.
- This time slot could have been allocated to some other station willing to send data.

- Effective bandwidth/bandwidth utilization/Throughput
= Efficiency * Bandwidth
- Maximum available effective bandwidth
= Total no of stations* Bandwidth requirement of 1 station.

Problem:

If transmission delay and propagation delay of a packet in Time division multiplexing is 1 msec each at 8Mbps bandwidth, then-

- a. Find the efficiency?
- b. Find the effective bandwidth/throughput?
- c. How many maximum stations can be connected to the network if each station requires 4kbps bandwidth ?

In a TDM medium access control bus LAN, each station is assigned one time slot per cycle for transmission. Assume that the length of each time slot is the time to transmit 100 bits plus the end-to-end propagation delay. Assume a propagation speed of $2 \times 10^8 m/sec$. The length of the LAN is 1 km with a bandwidth of 10 Mbps. The maximum number of stations that can be allowed in the LAN so that the throughput of each station can be 2/3 Mbps is

- A. 3
- B. 5
- C. 10
- D. 20

[GATE IT – 2005]

THANK YOU!

Here's to a cracking journey ahead!